

Package ‘elements’

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Title Ecological Niche Models for the Plants of Europe

Description A R package containing Ecological Niche Models (ENMs) for the most prevalent plants, bryophytes and terricolous lichens in the European Vegetation Archive (EVA).

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License GPL (>= 3)

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LazyDataCompression xz

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Suggests knitr,
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URL <https://github.com/NERC-CEH/elements>, <https://nerc-ceh.github.io/elements/>

BugReports <https://github.com/NERC-CEH/elements/issues>

Depends R (>= 4.4)

LazyData true

VignetteBuilder knitr

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ALEData	<i>ALE data</i>
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Description

A dataset containing the Accumulated Local Effect (ALE) (Molnar, 2018) data for the models, generated using 1000 samples from the training data.

Usage

ALEData

Format

A data frame with 1514450 rows and 4 columns, the definitions of which are:

taxon_code The taxon, see `elements::TaxonomicBackbone`.

x The variable value.

y The ALE value.

variable The variable name.

Details

ALEData

env_filter	<i>Retrieve the most suitable taxa for a given set of environmental variable values</i>
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Description

Retrieve the most suitable taxa for a given set of environmental variable values supplied in the 'predictors' argument. To sets of methods are available:

1. "svm" which generates predictions using `elements::predict_occ` and uses the resultant probability values;
2. "mean" and "median" which calculates the scaled euclidean distance between the values supplied in the 'predictors' argument and the mean or median niche positions as present in `elements::NicheWidths`.

Usage

```
env_filter(
  predictors = elements::ExampleScenarios[1, ],
  taxa = elements::TaxonomicBackbone$taxon_code,
  method = "svm",
  limit = NULL,
  exclude = NULL,
  threshold = NULL,
  toptaxa = NULL
)
```

Arguments

predictors	A data frame of predictors. Must include atleast one the following columns: L, M, N, R, S, SD, GP, bio05, bio06, bio16, bio17. Columns not included must then be included in the 'exclude' argument.
taxa	A vector of strings containing one or more taxa to generate predictions for.
method	One of "svm", "mean", or "median".
limit	A string representing the niche width quantiles, one of "min_max", "q01_q99", "q05_q95", "q25_q75". Which if set assigns a probability of 0 to a set of predictors if one or more of those predictors are outside the stipulated quantile ranges. Only applied if pa = "Present". Optional.
exclude	Model variables to exclude from the distance calculation; passed to the 'holdout' argument of 'elements::predict_occ' if the 'method' argument is "svm", otherwise when the 'method' argument is set to "mean" or "median" those variables are removed from the distance calculation.
threshold	A probability threshold to use as a cut off in the environmental filter. Only applicable when 'method' = "svm".
toptaxa	The number of top taxa, as defined by their probability when 'method' = "svm" or euclidean distance when 'method' = "mean" or "median".

Details

The svm method will produce more accurate results as it considers the position of the environmental variable values in the 11-dimensional hypervolume; however, if there are a large number of taxa-predictor combinations the mean and median methods offer a faster alternative.

NOTE: The "mean" and "median" methods do not produce realistic results and so are currently included for demonstrative purposes only.

Value

A dataframe containing three columns: taxon_code, rank, and Present (if 'method' = "svm") or Distance (if 'method' = "mean" or "median").

Examples

```
elements::startup(); elements::env_filter(predictors = elements::ExampleScenarios[1,], taxa = elements::TaxonomicBackbone$taxon_code)
```

ExampleData1

*Example predictor data***Description**

A dataset containing a randomised sample of 100 presences and 100 absences from the training and test data for one taxon: *Stellaria graminea* (*stellaria_graminea*)

Usage

ExampleData1

Format

A data frame with 200 rows and 11 columns, the definitions of which are:

L Light**M** Soil Moisture**N** Soil Nitrogen**R** Reaction**S** Salinity**SD** Soil Disturbance**GP** Grazing Pressure**bio05** Maximum temperature in the warmest month**bio06** Minimum temperature in the coldest month**bio16** Precipitation in the wettest quarter**bio17** Precipitation in the driest quarter**Details**

ExampleData1

References

Copernicus Climate Change Service, 2021. Downscaled bioclimatic indicators for selected regions from 1950 to 2100 derived from climate projections. <https://doi.org/10.24381/CDS.0AB27596>

Dengler, J., Jansen, F., Chusova, O., Hüllbusch, E., Nobis, M.P., Meerbeek, K.V., Axmanová, I., Bruun, H.H., Chytrý, M., Guarino, R., Karrer, G., Moeys, K., Raus, T., Steinbauer, M.J., Tichý, L., Tyler, T., Batsatsashvili, K., Bitá-Nicolae, C., Didukh, Y., Diekmann, M., Englisch, T., Fernández-Pascual, E., Frank, D., Graf, U., Hájek, M., Jelaska, S.D., Jiménez-Alfaro, B., Julve, P., Nakhutsrishvili, G., Ozinga, W.A., Ruprecht, E.-K., Šilc, U., Theurillat, J.-P., Gillet, F., 2023. Ecological Indicator Values for Europe (EIVE) 1.0. Vegetation Classification and Survey 4, 7–29. <https://doi.org/10.3897/VCS.98324>

Midolo, G., Herben, T., Axmanová, I., Marcenò, C., Pätzsch, R., Bruehlheide, H., Karger, D.N., Ačić, S., Bergamini, A., Bergmeier, E., Biurrun, I., Bonari, G., Čarni, A., Chiarucci, A., De Sanctis, M., Demina, O., Dengler, J., Dziuba, T., Fanelli, G., Garbolino, E., Giusso del Galdo, G., Goral, F., Güler, B., Hinojos-Mendoza, G., Jansen, F., Jiménez-Alfaro, B., Lengyel, A., Lenoir, J., Pérez-Haase, A., Piech, R., Prokhorov, V., Rašomavičius, V., Ruprecht, E., Rüşni, S., Šilc, U., Škvorc,

Ž., Stančić, Z., Tatarenko, I., Chytrý, M., 2023. Disturbance indicator values for European plants. *Global Ecology and Biogeography* 32, 24–34. <https://doi.org/10.1111/geb.13603>

Tichý, L., Axmanová, I., Dengler, J., Guarino, R., Jansen, F., Midolo, G., Nobis, M.P., Van Meerbeek, K., Aćić, S., Attorre, F., Bergmeier, E., Biurrun, I., Bonari, G., Bruelheide, H., Campos, J.A., Čarni, A., Chiarucci, A., Čuk, M., Čušterevska, R., Didukh, Y., Dítě, D., Dítě, Z., Dziuba, T., Fanelli, G., Fernández-Pascual, E., Garbolino, E., Gavilán, R.G., Gégout, J.-C., Graf, U., Güler, B., Hájek, M., Hennekens, S.M., Jandt, U., Jašková, A., Jiménez-Alfaro, B., Julve, P., Kambach, S., Karger, D.N., Karrer, G., Kavgacı, A., Knollová, I., Kuzemko, A., Küzmič, F., Landucci, F., Lengyel, A., Lenoir, J., Marcenò, C., Moeslund, J.E., Novák, P., Pérez-Haase, A., Peterka, T., Pielech, R., Pignatti, A., Rašomavičius, V., Rūsiņa, S., Saatkamp, A., Šilc, U., Škvorc, Ž., Theurillat, J.-P., Wohlgemuth, T., Chytrý, M., 2023. Ellenberg-type indicator values for European vascular plant species. *Journal of Vegetation Science* 34, e13168. <https://doi.org/10.1111/jvs.13168>

ExampleData2

Example predictor data

Description

A dataset containing a randomised sample of 100 presences and 100 absences from the training and test data for two taxa: *Stellaria graminea* (*stellaria_graminea*) and *Silene flos-cuculi* (*silene_flos-cuculi*).

Usage

ExampleData2

Format

A data frame with 400 rows and 12 columns, the definitions of which are:

L Light

M Soil Moisture

N Soil Nitrogen

R Reaction

S Salinity

SD Soil Disturbance

GP Grazing Pressure

bio05 Maximum temperature in the warmest month

bio06 Minimum temperature in the coldest month

bio16 Precipitation in the wettest quarter

bio17 Precipitation in the driest quarter

taxon_code The taxon, see `elements::TaxonomicBackbone`.

Details

ExampleData2

References

- Copernicus Climate Change Service, 2021. Downscaled bioclimatic indicators for selected regions from 1950 to 2100 derived from climate projections. <https://doi.org/10.24381/CDS.0AB27596>
- Dengler, J., Jansen, F., Chusova, O., Hüllbusch, E., Nobis, M.P., Meerbeek, K.V., Axmanová, I., Bruun, H.H., Chytrý, M., Guarino, R., Karrer, G., Moeys, K., Raus, T., Steinbauer, M.J., Tichý, L., Tyler, T., Batsatsashvili, K., Bitá-Nicolae, C., Didukh, Y., Diekmann, M., Englisch, T., Fernández-Pascual, E., Frank, D., Graf, U., Hájek, M., Jelaska, S.D., Jiménez-Alfaro, B., Julve, P., Nakhutsrishvili, G., Ozinga, W.A., Ruprecht, E.-K., Šilc, U., Theurillat, J.-P., Gillet, F., 2023. Ecological Indicator Values for Europe (EIVE) 1.0. Vegetation Classification and Survey 4, 7–29. <https://doi.org/10.3897/VCS.98324>
- Midolo, G., Herben, T., Axmanová, I., Marcenò, C., Pätsch, R., Bruelheide, H., Karger, D.N., Ačić, S., Bergamini, A., Bergmeier, E., Biurrun, I., Bonari, G., Čarni, A., Chiarucci, A., De Sanctis, M., Demina, O., Dengler, J., Dziuba, T., Fanelli, G., Garbolino, E., Giusso del Galdo, G., Goral, F., Güler, B., Hinojos-Mendoza, G., Jansen, F., Jiménez-Alfaro, B., Lengyel, A., Lenoir, J., Pérez-Haase, A., Pielech, R., Prokhorov, V., Rašomavičius, V., Ruprecht, E., Rūsiņa, S., Šilc, U., Škvorc, Ž., Stančić, Z., Tatarenko, I., Chytrý, M., 2023. Disturbance indicator values for European plants. *Global Ecology and Biogeography* 32, 24–34. <https://doi.org/10.1111/geb.13603>
- Tichý, L., Axmanová, I., Dengler, J., Guarino, R., Jansen, F., Midolo, G., Nobis, M.P., Van Meerbeek, K., Ačić, S., Attorre, F., Bergmeier, E., Biurrun, I., Bonari, G., Bruelheide, H., Campos, J.A., Čarni, A., Chiarucci, A., Čuk, M., Čušterevska, R., Didukh, Y., Dítě, D., Dítě, Z., Dziuba, T., Fanelli, G., Fernández-Pascual, E., Garbolino, E., Gavilán, R.G., Gégout, J.-C., Graf, U., Güler, B., Hájek, M., Hennekens, S.M., Jandt, U., Jašková, A., Jiménez-Alfaro, B., Julve, P., Kambach, S., Karger, D.N., Karrer, G., Kavgacı, A., Knollová, I., Kuzemko, A., Küzmič, F., Landucci, F., Lengyel, A., Lenoir, J., Marcenò, C., Moeslund, J.E., Novák, P., Pérez-Haase, A., Peterka, T., Pielech, R., Pignatti, A., Rašomavičius, V., Rūsiņa, S., Saatkamp, A., Šilc, U., Škvorc, Ž., Theurillat, J.-P., Wohlgemuth, T., Chytrý, M., 2023. Ellenberg-type indicator values for European vascular plant species. *Journal of Vegetation Science* 34, e13168. <https://doi.org/10.1111/jvs.13168>

ExamplePlot

The EVA plot used to form the example scenarios

Description

A dataset containing the taxon presences and percentage cover for the EVA plot used to form the example scenarios present in `elements::ExampleScenarios`.

Usage

ExamplePlot

Format

A data frame with 6 rows and 3 columns, the definitions of which are:

taxon_name The scientific names of the taxa present in the example plot.

taxon_code The taxon codes for the taxon_name values, see `elements::TaxonomicBackbone`.

cover_perc The percentage cover of the taxa present in example plot.

Details

ExamplePlot

ExampleScenarios

*Predictor data for three example scenarios***Description**

A dataset containing predictor data for five example scenarios, in three groups: (A) Climate Change - RCP4.5 and (A) Climate Change - RCP8.5, (B) Grazing Intensification and (B) Grazing Reduction, and (C) Nutrient Enrichment.

Usage

ExampleScenarios

Format

A data frame with 26 rows and 14 columns, the definitions of which are:

L Light**M** Soil Moisture**N** Soil Nitrogen**R** Reaction**S** Salinity**SD** Soil Disturbance**GP** Grazing Pressure**bio05** Maximum temperature in the warmest month**bio06** Minimum temperature in the coldest month**bio16** Precipitation in the wettest quarter**bio17** Precipitation in the driest quarter**scenario** The scenario name.**timeslice** The scenario timeslice, either a year or period.**scenario_code** The scenario code: a, b, or c.**Details**

ExampleScenarios

Imbalances	<i>Presence-Absence Imbalances</i>
Description The number of presences and absences in the EVA, suitable samples within the EVA, and the training and test data, along with the presence-absence and absence-presence imbalance ratios. Usage Imbalances Format A data frame with 18102 rows and 6 columns, the definitions of which are: Stage The plot selection 'stage'. Absent The number of absences. Present The number of presences. PA_Imbalance The presence-absence imbalance ratio. AP_Imbalance The absence-presence imbalance ratio. taxon_code The taxon code, see <code>elements::TaxonomicBackbone</code> Details Imbalances	
NicheWidths	<i>Niche width data</i>

Description

A dataset Niche width data for the modelled taxa.

Usage

NicheWidths

Format

A data frame with 66374 rows and 14, the definitions of which are:

variable The variable name.

mean The mean variable value.

min The minimum variable value.

max The maximum variable value.

median The median variable value.

q01 The 1% quantile value.

q05 The 5% quantile value.

q10 The 10% quantile value.

q25 The 25% quantile value.

q75 The 75% quantile value.

q90 The 90% quantile value.

q95 The 95% quantile value.

q99 The 99% quantile value.

taxon_code The taxon, see `elements::TaxonomicBackbone`.

Details

NicheWidths

PDPData

PDP data

Description

A dataset containing the Partial Dependency Profile (PDP) (Molnar, 2018) data for the models, generated using 1000 samples from the training data.

Usage

PDPData

Format

A data frame with 1458820 rows and 4 columns, the definitions of which are:

taxon_code The taxon, see `elements::TaxonomicBackbone`.

x The variable value.

y The PDP value.

variable The variable name.

Details

PDPData

PerformanceMeasures	<i>Model performance measures</i>
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Description

A dataset containing a number of performance measures for each model.

Usage

PerformanceMeasures

Format

A data frame with 6631 rows and 8 columns, the definitions of which are:

taxon_code The taxon, see `elements::TaxonomicBackbone`.

Holdout.PrecisionRecallAreaUnderCurve The Precision-Recall Area Under the Curve (PRAUC), calculated using the random holdout sample test data.

Holdout.Precision The Precision, calculated using the random holdout sample test data.

Holdout.Recall The Recall, calculated using the random holdout sample test data.

Holdout.Sensitivity The Sensitivity, calculated using the random holdout sample test data.

Holdout.Specifity The Specificity, calculated using the random holdout sample test data.

Holdout.BalancedAccuracy The Balanced Accuracy, calculated using the random holdout sample test data.

STCV.BalancedAccuracy The Balanced Accuracy, calculated during the modelling fitting process which used spatio-temporal 10-fold cross-validation.

Details

PerformanceMeasures

plot_me	<i>Plot the marginal effect values for one or more taxa and set of variables</i>
---------	--

Description

Plot the Accumulated Local Effect (ALE) or Partial Dependence Profile (PDP) (Molnar, 2018; Molnar, 2022) marginal effect curves for one or more taxa and a set of variables.

Usage

```
plot_me(
  taxa,
  me_type = "pdp",
  free_y = TRUE,
  presences = TRUE,
  eivs = TRUE,
  normalise = TRUE,
  vars = c("L", "M", "N", "R", "S", "SD", "GP", "bio05", "bio06", "bio16", "bio17"),
  lmw = 15,
  lts = 0.75
)
```

Arguments

taxa	A vector of one or more <code>taxon_code</code> strings, see <code>elements::TaxonomicBackbone</code> .
me_type	A string representing the marginal effect plot type, one of "ale" or "pdp".
free_y	A boolean. If TRUE the Y axis scales are independent and free for all subplots. If FALSE the Y axis scales are fixed between all subplots.
presences	A boolean. If TRUE a box and whiskers plot showing the distribution of presences along each variable will be displayed.
eivs	A boolean. If TRUE a point representing the EIV value and arrows representing the EIV niche widths for the taxon will be displayed, where available in <code>elements::VariableData</code> .
normalise	A boolean. If TRUE and <code>me_type == "pdp"</code> the y axes are normalised using min-max re-scaling.
vars	A vector of variables. Must include atleast one of the following columns: "L", "M", "N", "R", "S", "SD", "GP", "bio05", "bio06", "bio16", and "bio17".
lmw	The width of the outer margin containing the legend, passed to the "oma" argument of <code>graphics::par</code> . Adjust to ensure the legend is given enough room.
lts	The size of the legend text, passed to the "cex" argument of <code>graphics::legend</code> . Adjust to ensure the legend text size is appropriate.

Details

If the number of taxa is one, setting the 'presences' argument to TRUE a box and whiskers plot showing the distribution of presences is overlaid and by setting the 'eivs' argument to TRUE a point and arrows showing the EIV and niche width values are overlaid, where available in `elements::VariableData`.

The presence-absence imbalance in the training data varies by taxon. This 'ghost of imbalance' (Jiménez-Valverde & Lobo, 2006) has several impacts on the PDP plots:

1. the optimum value of the PDP curve may be less than 1.
2. the entire PDP curve may sit below $y = 0.5$ (the presence-absence threshold).

When inspecting the PDP plots, it is therefore important to pay more attention to the shape of the response, rather than the absolute PDP value. However, by setting the `normalise` argument to TRUE, the PDP plot data is transformed using min-max normalisation/re-scaling.

In some instances the ALE curves may reflect 'inverted' responses which are not ecologically realistic, this is most often seen in situations where the distribution of presences along a variable gradient is extremely narrow and/or where there is a non-unimodal distribution, which causes extrapolation

issues in the ALE calculations. For example, *Gymnocarpium robertianum* has a extremely narrow distribution of plot-mean S values, with a maximum value of 1. In these instances it is important to also visualise the PDP plots, which should then be prioritised when inspecting the shape of the univariate response.

Value

A composite plot showing the marginal effects and optionally the distribution of presences for selected model variables.

References

- Jiménez-Valverde, A., Lobo, J.M., 2006. The ghost of unbalanced species distribution data in geographical model predictions. *Diversity and Distributions* 12, 521–524. <https://doi.org/10.1111/j.1366-9516.2006.00267.x>
- Molnar, C., 2018. iml: An R package for Interpretable Machine Learning. *Journal of Open Source Software* 3, 786. <https://doi.org/10.21105/joss.00786>
- Molnar, C., 2022. *Interpretable Machine Learning: A Guide For Making Black Box Models Explainable*. Independently published, Munich, Germany.

Examples

```
elements::plot_me(taxa = "ajuga_reptans", me_type = "ale", free_y = FALSE, presences = TRUE, eivs = TRUE, normalise = FALSE)

elements::plot_me(taxa = c("galium_boreale", "galium_sylvaticum", "galium_uliginosum"), me_type = "ale", free_y = FALSE, presences = TRUE, eivs = TRUE, normalise = FALSE)
```

predict_occ	<i>Generate predictions for the probability of occurrence for specified taxa</i>
-------------	--

Description

Generate predictions for the probability of occurrence (presence and/or absence) using a data frame of predictors for a one or more taxa specified in either:

1. the 'taxa_codes' argument, or
2. a column in the predictors data frame named 'taxon_code'

Usage

```
predict_occ(
  taxa_codes,
  predictors,
  pa = "Present",
  limit = NULL,
  holdopt = NULL,
  dp = 3,
  append_predictors = TRUE
)
```

Arguments

taxa_codes	A vector of strings containing one or more taxa to generate predictions for. Optional.
predictors	A data frame of predictors. Must include the following columns: L, M, N, R, S, SD, GP, bio05, bio06, bio16, bio17, and taxon
pa	One of "Present", "Absent", or c("Present", "Absent").
limit	A string representing the niche width quantiles, one of "min_max", "q01_q99", "q05_q95", "q25_q75". Which if set assigns a probability of 0 to a set of predictors if one or more of those predictors are outside the stipulated quantile ranges. Only applied if pa = "Present". Optional.
holdopt	Hold one or more variables at their optimum values. NULL by default, else a vector of variable codes, e.g. c("SD", "GP").
dp	The number of decimal places to round the probability values to.
append_predictors	A boolean. If TRUE return the predictors data frame with the results in an additional column.

Details

A number of optional arguments provide additional control over the use of the models. First, using the 'limit' argument the probabilities may be set to 0 if one or more of the predictor values are outside a stipulated quantile range, e.g. the 1% and 99% quantiles by setting the 'limit' argument to "q01_q99". This may be used to strictly enforce the assignment of probability values of 0. As the quantile values present in `elements::NicheWidths` are derived using all presences of each taxon in the EVA, they may better represent the upper and lower tolerances of a taxons ecological niche, especially for taxa where the presences were undersampled. Second, using the 'holdopt' argument one or more predictor variables can be held constant, e.g. to hold soil disturbance constant supply c("SD") to the 'holdopt' argument. This may be useful in instances where you wish to remove the influence of one or more variables on the model results.

If taxon codes are supplied in the 'taxa_codes' argument and there is a column in the predictors data frame named 'taxon_code', the taxa present in the 'taxa_codes' argument will be used and results will be calculated using the entire predictors data frame and the 'taxon_code' column will be replaced.

NOTE: to use this function you must first run `elements::startup()`

Value

A data frame containing the probability of occurrence (Present and/or Absent) for each taxon, if specified, appended to the predictorsS data.

Examples

```
## Not run:
elements::startup()
```

```
# Generate predictions using a data frame containing taxa_codes in the 'taxon' column.
elements::predict_occ(taxa_codes = NULL, predictors = elements::ExampleData2, pa = "Present", limit = NULL, ho
```

```
# Generate predictions for taxa specified in the 'taxa_codes' argument using a data frame containing only predic
elements::predict_occ(taxa_codes = c("stellaria_graminea", "silene_flos-cuculi"), predictors = elements::Exar
```

```
# Generate predictions for taxa specified in the 'taxa_codes' argument using a data frame containing both predictors and taxonomic
elements::predict_occ(taxa_codes = c("stellaria_graminea", "silene_flos-cuculi"), predictors = elements::ExaminedData, taxa_codes = elements::TaxonomicBackbone)

elements::shutdown()

## End(Not run)
```

predict_occ_taxon	<i>Generate predictions for the probability of occurrence</i>
-------------------	---

Description

Generate predictions for the probability of occurrence (presence and/or absence) for a given taxon.

Usage

```
predict_occ_taxon(
  taxon,
  predictors,
  pa = "Present",
  limit = NULL,
  holdopt = NULL,
  dp = 3,
  append_predictors = TRUE
)
```

Arguments

taxon	The taxon_code, see elements::TaxonomicBackbone.
predictors	A data frame of predictors. Must include the following columns: L, M, N, R, S, SD, GP, bio05, bio06, bio16, and bio17
pa	One of "Present", "Absent", or c("Present", "Absent").
limit	A string representing the niche width quantiles, one of "min_max", "q01_q99", "q05_q95", "q10_q90", "q25_q75". Which if set assigns a probability of 0 to the Present column and/or 1 to the Absent column to a set of predictors if one or more of those predictors are outside the stipulated quantile ranges. Optional.
holdopt	Hold one or more variables at their optimum values. NULL by default, else a vector of variable codes, e.g. c("SD", "GP").
dp	The number of decimal places to round the probability values to.
append_predictors	A boolean. If TRUE return the predictors data frame with the results in an additional column.

Details

A number of optional arguments provide additional control over the use of the models. First, using the 'limit' argument the probabilities may be set to 0 if one or more of the predictor values are outside a stipulated quantile range, e.g. the 1% and 99% quantiles by setting the 'limit' argument to "q01_q99". This may be used to strictly enforce the assignment of probability values of 0. As the quantile values present in elements::NicheWidths are derived using all presences of each taxon

in the EVA, they may better represent the upper and lower tolerances of a taxons ecological niche, especially for taxa where the presences were undersampled. Second, using the 'holdopt' argument one or more predictor variables can be held constant, e.g. to hold soil disturbance constant supply c("SD") to the 'holdopt' argument. This may be useful in instances where you wish to remove the influence of one or more variables on the model results.

NOTE: to use this function you must first run elements::startup()

Value

A data frame containing the probability of occurrence (Present and/or Absent), if specified, appended to the predictors data.

Examples

```
## Not run:
elements::startup()
elements::predict_occ_taxon(taxon = "stellaria_graminea", predictors = elements::ExampleData1, pa = "Present")
elements::shutdown()

## End(Not run)
```

shutdown	<i>Remove the filehash database connection from the global environment</i>
----------	--

Description

Remove the filehash database connection from the global environment

Usage

```
shutdown()
```

Examples

```
elements::shutdown()
```

startup	<i>Initiliase a connection to the filehash database containing the Ecological Niche Models (ENMs)</i>
---------	---

Description

Initiliase a connection to the filehash database containing the Ecological Niche Models (ENMs)

Usage

```
startup(models = NULL)
```

Arguments

models One of "all", "test", or NULL, indicating whether the complete set of models should be loaded, or the test set of models. NULL by default, which loads all models.

Value

A connection to the filehash database containing the ENMs in an environment named `elementsEnv` within the parent environment.

Examples

```
elements::startup()
```

TaxonomicBackbone	<i>Taxonomic backbone</i>
-------------------	---------------------------

Description

The taxon names and codes for the modeled taxa in the EVA, with the associated taxon concept information retrieved from GBIF.

Usage

```
TaxonomicBackbone
```

Format

A data frame with 6631 rows and 13 columns, the definitions of which are:

taxon_name The taxon names for the modelled taxa.

taxon_code The taxon codes used throughout the package, formed from the `taxon_name` values by coercing all letters to lower and replacing whitespace with underscores.

scientificName The full scientific name of the taxon, which includes the author. Retrieved from GBIF.

canonicalName The full name of the taxon. Retrieved from GBIF.

species The accepted name for the taxon, following GBIF.

genus The parent Genus taxon associated with the Species.

family The parent Family taxon associated with the Species.

order The parent Order taxon associated with the Species.

class The parent Class taxon associated with the Species.

phylum The parent Phylum taxon associated with the Species.

kingdom The parent Kingdom taxon associated with the Species.

Details

```
TaxonomicBackbone
```

VariableData

*Environmental indicator value data***Description**

A dataset containing a combined dataset of Environmental Indicator Values (EIVs) from Dengler et al (2023), Midolo et al (2023), and Tichý et al (2023).

Usage

VariableData

Format

A data frame with 9406 rows and 26 columns, the definitions of which are:

DF Disturbance frequency (Midolo et al., 2023)

DF.sd Disturbance frequency standard deviation (Midolo et al., 2023)

DFH.sd Disturbance frequency herb layer standard deviation (Midolo et al., 2023)

DFh Disturbance frequency herb layer (Midolo et al., 2023)

DS Disturbance severity (Midolo et al., 2023)

DS.sd Disturbance severity standard deviation (Midolo et al., 2023)

DSH.sd Disturbance severity herb layer standard deviation (Midolo et al., 2023)

DSH Disturbance severity herb layer (Midolo et al., 2023)

GP Grazing pressure (Midolo et al., 2023)

GP.sd Grazing pressure standard deviation (Midolo et al., 2023)

L Light (Dengler et al., 2023)

L.nw Light niche width (Dengler et al., 2023)

M Moisture (Dengler et al., 2023)

M.nw Moisture niche width (Dengler et al., 2023)

MF Mowing frequency (Midolo et al., 2023)

MF.sd Mowing frequency standard deviation (Midolo et al., 2023)

N Nitrogen (Dengler et al., 2023)

N.nw Nitrogen niche width (Dengler et al., 2023)

R Reaction (Dengler et al., 2023)

R.nw Reaction niche width (Dengler et al., 2023)

S Salinity (Tichý et al 2023)

SD Soil disturbance (Midolo et al., 2023)

SD.sd Soil disturbance standard deviation (Midolo et al., 2023)

T Temperature (Dengler et al., 2023)

T.nw Temperature niche width (Dengler et al., 2023)

taxon_name The taxon name.

Details

VariableData

References

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- Dengler, J., Jansen, F., Chusova, O., Hüllbusch, E., Nobis, M.P., Meerbeek, K.V., Axmanová, I., Bruun, H.H., Chytrý, M., Guarino, R., Karrer, G., Moeys, K., Raus, T., Steinbauer, M.J., Tichý, L., Tyler, T., Batsatsashvili, K., Bitá-Nicolae, C., Didukh, Y., Diekmann, M., Englisch, T., Fernández-Pascual, E., Frank, D., Graf, U., Hájek, M., Jelaska, S.D., Jiménez-Alfaro, B., Julve, P., Nakhutsrishvili, G., Ozinga, W.A., Ruprecht, E.-K., Šilc, U., Theurillat, J.-P., Gillet, F., 2023. Ecological Indicator Values for Europe (EIVE) 1.0. Vegetation Classification and Survey 4, 7–29. <https://doi.org/10.3897/VCS.98324>
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VariableLookup

Environmental variable names and codes

Description

A dataset containing a lookup for the Ecological Indicator Value (EIV) and bioclimatic variable codes used throughout the `elements` package and full variable names.

Usage

VariableLookup

Format

A data frame with 29 rows and 5 columns, the definitions of which are:

raw_name The raw name of the variable as present in the parent dataset.

variable_code The variable code, used throughout the `elements` R package.

variable_name A 'tidy' variable name without spaces.
variable_plot_name A 'tidy' variable name with spaces.
model_var A boolean indicating whether the variable was included as a predictor.

Details

VariableLookup

VariableNames	<i>Model variables codes</i>
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Description

A vector containing the environmental variable codes for the model variables; equivalent to the variable_code values in the elements::VariableLookup data frame for rows where model_var is TRUE.

Usage

VariableNames

Format

A vector containing strings.

Details

VariableNames

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